

BUS 393
FINAL WRITE UP PART A
AIRLINE BOOKING PLATFORM

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Scope:

The domain chosen for this project is an airline booking platform, which is a centralized system that allows customers to search for flights, make reservations, assign passengers, select seats, add luggage, and complete payments. The platform involves key entities within air travel, such as customers, airlines, airports, routes, flights, bookings, passengers, and payments.

The specific aspect of the business focused on for the modeling is the flight booking and reservation process. The database focuses on managing the complete lifecycle of a booking. In the context of an airline booking platform, customers are able to create a reservation which in turn leads to passenger registration, ticket assignment, seat selection, luggage purchases, and payment processing. Additionally, this model displays the relationships between customers, bookings, flights, and passengers while enforcing business rules such as one payment per booking, passenger subtype classifications (adult, minor, or employee), and flight-route-airline relationships. By concentrating on reservation management, the database supports accurate transaction processing, passenger tracking, revenue analysis, and operational reporting for airline booking activities.

Business Rules

Our project is centered around two categories of business rules: Transactional Rules and Entity Rules. Transactional rules govern how booking references function within an airline booking system, while entity rules govern how customers, passengers, flights, and routes interact within the airline domain.

TRANSACTIONAL RULES:

- Each booking generates exactly one payment record, ensuring that every reservation has a corresponding financial transaction.
- A booking may contain multiple passengers and multiple flights. This many-to-many relationship is resolved through the BOOKING_FLIGHT associative entity, where each record represents a single ticket.
- Each BOOKING_FLIGHT record must reference exactly one booking, one flight, and one passenger, ensuring that every ticket is uniquely associated with a specific traveler on a specific flight within a specific reservation.

- A BOOKING_FLIGHT record may have multiple luggage add-ons and seat-related records associated with it, allowing ancillary services to be tracked at the individual ticket level.

ENTITY RULES:

- A customer can make many bookings but each booking belongs to only one customer
- A customer may refer to multiple other customers, while each customer may be referred by at most one existing customer. This relationship is implemented through a unary self-reference within the CUSTOMER entity.
- Each flight must be operated by one airline and assigned to one route, ensuring clear ownership of the flight and a defined origin-destination path.
- Every route consists of exactly one origin airport and one destination airport, while airports may participate in many routes as either origins or destinations.
- Every passenger must belong to exactly one disjoint subtype: Adult Passenger, Minor Passenger, or Employee Passenger. A passenger cannot simultaneously belong to more than one subtype.
- A minor passenger must have a guardian who is an adult passenger, ensuring that all minor travelers are associated with a responsible adult.
- Employee passengers represent airline personnel and therefore cannot be associated with a customer account.

Definitions: Each entity is defined below in business terms, with notes on any attributes that are not self-explanatory.

- A *customer* is a person who registers on the platform to search for and book flights. Customers provide contact information and can refer other customers through the platform's referral program. Attributes: customer_id (PK), first_name, last_name, email, phone, referred_by (FK, a unary self-reference to customer_id; NULL means the customer was not referred by anyone).
- An *airline* is a company registered on the platform that operates one or more flights. Each airline is identified by a unique code and name. Attributes: airline_id (PK), airline_name, airline_code.
- An *airport* is a physical location that serves as either the origin or the destination of a route. The same airport can appear in many routes in either role. Attributes: airport_id (PK), airport_code, airport_name, city, state.
- A *route* is a pairing of an origin airport and a destination airport. It captures the physical path a flight travels and stores the distance between the two airports. The same airport can serve as the origin or destination of many routes. Attributes: route_id (PK), origin_airport_id (FK), destination_airport_id (FK), distance_miles

- A *flight* is a specific scheduled trip operated by an airline along a route. It captures departure and arrival times along with a base price. One airline can operate many flights and one route can have many flights scheduled on it. Attributes: flight_id (PK), airline_id (FK), route_id (FK), flight_number, departure_time, arrival_time, base_price
- A *booking* is a reservation transaction made by a customer on the platform. It records when the reservation was made and its current status. One customer can make many bookings but each booking belongs to exactly one customer. Attributes: booking_id (PK), customer_id (FK), booking_date, booking_status (the current state of the reservation, e.g. Confirmed, Pending, or Cancelled)
- A *passenger* is any person traveling on a flight. Every passenger must belong to exactly one of three disjoint subtypes: adult, minor, or employee. Paying passengers are linked to a customer but employee passengers are not. Attributes: passenger_id (PK), customer_id (FK - nullable), first_name, last_name, passport_number, passenger_type (constrained to one of three values: adult, minor, or employee)
- A *booking_flight* is a ternary associative entity that links exactly one booking, one flight, and one passenger; representing one ticket. It also serves as the anchor point for seat assignments and luggage add-ons. Attributes: booking_flight_id (PK), booking_id (FK), flight_id (FK), passenger_id (FK), ticket_price
- A payment records the financial transaction associated with a booking. Each booking generates exactly one payment which captures the method, amount, date, and status of the transaction. Attributes: payment_id (PK), booking_id (FK), payment_date, payment_method, amount_paid, payment_status
- A luggage record stores add-on baggage fees for a specific booking_flight. A seat assignment records the seat number, cabin class, and seat fee for a specific booking_flight.
- Luggage Attributes: luggage_id (PK), booking_flight_id (FK), luggage_type, quantity, fee
- Seat Assignment Attributes: seat_assignment_id (PK), booking_flight_id (FK), seat_number, cabin_class, seat_fee

E-R Diagram: The full entity-relationship diagram is shown below on a single page. Each relationship is labeled with a descriptor, and the disjoint passenger hierarchy is shown with a circle marked “d.”

